Server Deep Dive (With Descriptions): Recall, Ranking, Synonyms, Analytics, Admin APIs

Prefix Recall

Prefix recall is the first retrieval stage. It narrows down candidates quickly based on typed prefixes, ensuring high recall and fast lookups.
Purpose: fast lexical candidates starting with the typed prefix.

Indexing: trie, DAWG, or inverted index with prefix keys.

Process: normalize input, compute prefix key, fetch top K by popularity/freshness.

Freshness: hot entity delta lists for quick updates.

Latency: 5–15 ms typical.

Fuzzy Recall

Fuzzy recall adds tolerance for user mistakes such as typos, spelling variations, or missing characters, improving user experience by not missing relevant results.
Purpose: recover typos/near matches when prefix recall is insufficient.

Algorithms: Levenshtein automaton (edit distance 1–2), or n-gram overlap.

Safeguards: cap expansion; locale-aware normalization.

Latency: 5–20 ms depending on query expansion.

Semantic Recall

Semantic recall retrieves results based on meaning rather than exact characters, powered by embeddings and ANN search. It expands recall to cover conceptually similar queries.
Purpose: capture meaning-related results beyond exact prefix match.

Embeddings: ANN indices like HNSW/IVF for fast vector search.

Triggers: long queries, tail terms, or low lexical hits.

Latency: 10–25 ms; fallback to lexical if degraded.

Personalized Ranking

Personalized ranking re-orders candidates using behavioral signals, popularity, and relevance features. It ensures the most useful results appear first, tailored to each user.
Signals: exactness, recency, popularity, user affinity, business rules, diversity.

Model: linear scoring or LTR; online features from KV store.

Privacy: opt-in only; TTL on user features; regional residency compliance.

Output: final ranked list with explainable scores.

Synonyms & Rules

Synonyms and rules help tune recall and ranking. Synonyms expand queries with equivalent terms, while rules let admins control visibility, pinning, or boosting results for business needs.
Synonyms: tv <-> television, locale-aware; versioned sets per index.

Rules: pin, boost, demote, hide based on query, type, time, or tenant.

Conflict resolution: priority ordering; highest priority wins.

APIs: POST /v1/indices/{id}/synonyms, POST /v1/indices/{id}/rules

Analytics

Analytics capture query and usage data to measure performance, improve ranking models, and detect issues. Key KPIs include CTR, abandonment rate, and latency trends.
Events: QueryEvent (query, latency, results), ImpressionEvent (ranked results shown), SelectionEvent (clicked entity).

KPIs: CTR, abandonment, time-to-first-suggestion, latency distributions.

Usage: nightly jobs update popularity signals; dashboards for latency and cache health.

APIs: POST /v1/events/query, POST /v1/events/selection, GET /v1/analytics/reports

Admin APIs

Admin APIs provide lifecycle control over indices, synonyms, rules, and ingestion. They enable admins to configure, monitor, and secure the autocomplete system at scale.
Index lifecycle: create, patch, snapshot, rebuild.

Synonyms: manage synonym sets per index.

Rules: manage business pin/boost/demote rules.

Ingestion: upsert/delete entities, stream ingestion.

Security: API key management with quotas/audit.

APIs: POST /v1/indices, POST /v1/entities:batchUpsert, DELETE /v1/entities/{id}